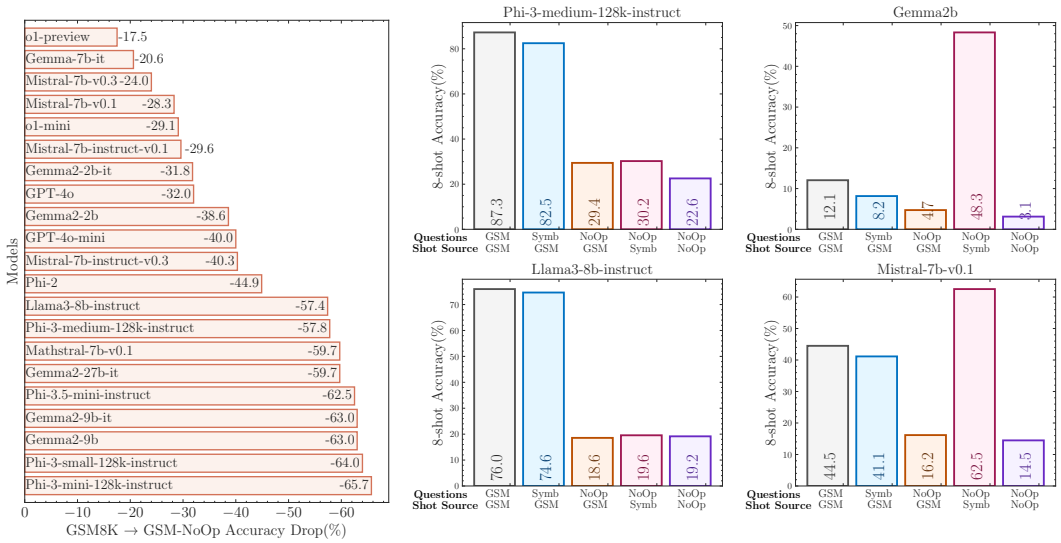

(a) (b) (c)

Figure 8: **(a)** The performance of models drops significantly on GSM−NoOp, with more recent models experiencing a greater decline than older ones. **(b)** As previously demonstrated, performance on GSM−Symbolic is very close to that on GSM8K. However, on GSM−NoOp, the significant drop in performance cannot be recovered, even when using the exact same question's variation as shots (NoOp−Symb) or when using different questions with different GSM−NoOpthat contain No-Op operations (NoOp−NoOp) as shots. **(c)** Notably, some models that perform significantly worse than those in (b) on GSM8K and GSM−Symbolic show much better performance on NoOp−Symb.

fact that the correct answer should ignore the No-Op statement. We observe that for the Llama-3-8B model, the performance remains the same compared to the original No-Op model, while for the Phi-3 model, performance slightly decreases.

5 Conclusion

In this work, we have investigated the reasoning capabilities of large language models (LLMs) and the limitations of current evaluations on GSM8K. We introduced GSM-Symbolic, a novel benchmark with multiple variants designed to provide deeper insights into the mathematical reasoning abilities of LLMs. Our extensive study reveals significant performance variability across different instantiations of the same question, challenging the reliability of current GSM8K results that rely on single-point accuracy metrics. We found that while LLMs exhibit some robustness to changes in proper names, they are more sensitive to variations in numerical values. We have also observed the performance of LLMs deteriorating as question complexity increases.

The introduction of GSM-NoOp exposes a critical flaw in LLMs' ability to genuinely understand mathematical concepts and discern relevant information for problem-solving. Adding seemingly relevant but ultimately inconsequential information to the logical reasoning of the problem led to substantial performance drops across models. Importantly, we demonstrate that LLMs struggle even when provided with multiple examples of the same question or examples containing similar irrelevant information. This suggests deeper issues in their reasoning processes that cannot be easily mitigated through few-shot learning or fine-tuning.

Ultimately, our work underscores significant limitations in LLMs' ability to perform genuine mathematical reasoning. The LLMs' high performance variance on different instances of the same question, their significant drop in performance with a slight increase in difficulty, and their sensitivity to inconsequential information indicate that their reasoning is fragile and may be more akin to sophisticated pattern matching rather than true logical reasoning. We believe further research is needed to develop AI systems capable of formal reasoning, moving beyond probabilistic pattern matching to achieve more robust and generalizable problem-solving skills. This remains a critical challenge for the field as we strive to create systems with human-like cognitive abilities or general intelligence.

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